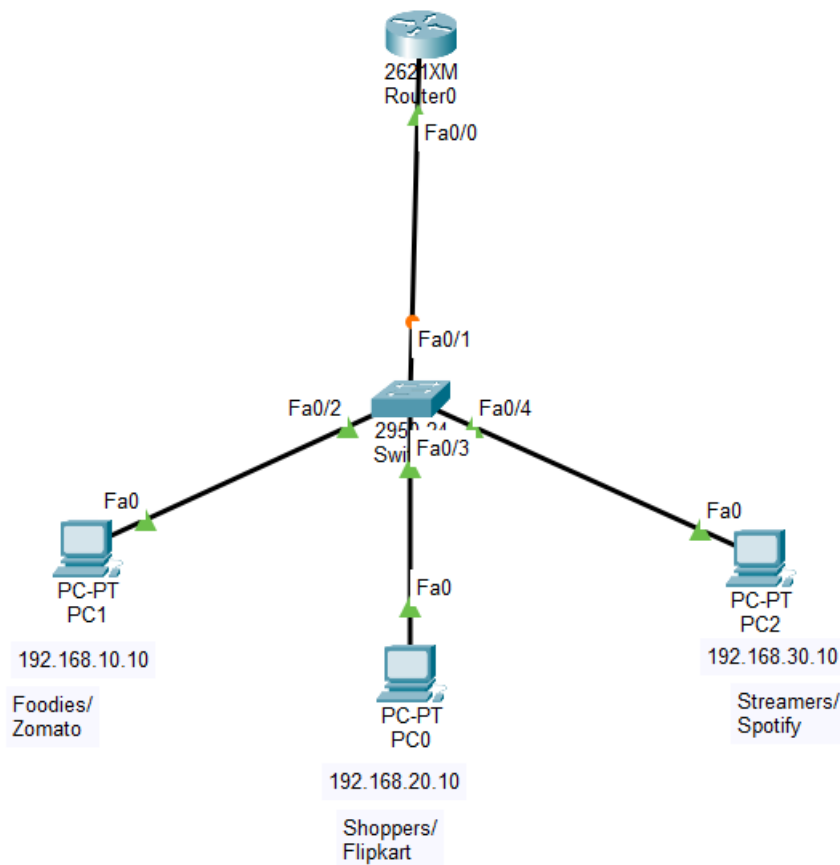


Inter-VLAN Routing Basics Theory: Methods of inter-VLAN routing: Router-on-a-stick, Layer 3 switches. Practical: Configure router-on-a-stick for inter-VLAN communication. Task: Troubleshoot inter-VLAN routing issues in a simulated scenario.

Task 1.

In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.



```

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Foodies
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Shoppers
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Streamers
Switch(config-vlan)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit

```

Task 2.

Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

```

Router(config)#end
Router#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet0/0.10	192.168.10.1	YES	manual	up	up
FastEthernet0/0.20	192.168.20.1	YES	manual	up	up
FastEthernet0/0.30	192.168.30.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0	unassigned	YES	unset	administratively down	down

```

Router#

```

Task 3.

Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

Device	VLAN	IP Address	Gateway
PC1	Foodies – 10	192.168.10.10	192.168.10.1
PC2	Shoppers – 20	192.168.20.10	192.168.20.1
PC3	Streamers – 30	192.168.30.10	192.168.30.1

```
Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
|
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Task 4.

Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity.

Hint: Check VLAN assignments, router sub-interface configurations, and trunk port settings.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
Switch#
Switch#show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Appliance trust: none
```

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=5ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

Task 5.

Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

Router-on-a-Stick	Layer 3 Switch
Uses one physical router interface with multiple sub-interfaces	Uses Switch Virtual Interfaces (SVIs) for each VLAN
Less expensive; can use an existing router	Faster inter-VLAN routing
Easy to configure and understand for small networks	Handles many VLANs and high traffic efficiently
One trunk link can become a bottleneck	Layer 3 switches are more expensive
Router interface can become a single point of congestion/failure	Configuration is slightly more advanced
Small networks and learning/lab environments	Medium and large networks

Router-on-a-Stick:

Advantages

- Less expensive if you already have a router.
- Simple and easy to understand for small networks.

Disadvantages

- All VLAN traffic shares one trunk link, creating a bottleneck.
- Router interface can become a bottleneck.

Layer 3 Switch :

Advantages

- Faster inter-VLAN routing because routing happens in the switch.
- can handle many VLANs efficiently.

Disadvantages

- Layer 3 switches are generally more expensive.
- Configuration can be more complex for beginners.